



REASONING

- If a man walks at the rate of 4 km/h, he misses a train by only 6 min. However, if he walks at the rate of 5 km/h he reaches the station 6 min before the arrival of the train. The distance covered by him to reach the station is
 - 4 km
 - 7 km
 - 9 km
 - 5 km
- The missing number in the given series 3, 6, 6, 12, 9, ..., 12 is
 - 15
 - 18
 - 11
 - 13
- A man runs 20 m towards east and turns right, runs 10 m and turns right, runs 9 m and turns left, runs 5 m and turns left, runs 12 m and finally turns left and runs, 6 m. Which direction is the man facing?
 - North
 - South
 - East
 - West
- In a club, there are certain number of males and females. If 15 females are absent, then number of males will be half of females. If 45 males are absent, then female strength will be 5 times that of males. Number of males actually present is
 - 45
 - 80
 - 105
 - 175
- The missing number in the following series 6, 12, 21, ..., 48 is
 - 40
 - 33
 - 38
 - 45

Directions (Q.Nos. 6- 8) Read the following passage carefully and answer the questions.

Six boys A, B, C, D, E and F are marching in a line. They are arranged according to their heights, the tallest being at the back and the shortest in the front. F is between B and A. E is shorter than D but taller than C who is taller than A. E and F have two boys between them. A is not shortest among them.

- Where is E?
 - Between A and B
 - Between C and A
 - Between D and C
 - In front of C
- If we start counting from the shortest, which boy is fourth in the line?
 - E
 - A
 - D
 - C
- Who is next to the shortest?
 - C
 - B
 - E
 - F

- Let x, y and z be distinct integers, x and y are odd and positive and z is even and positive. Which one of the following statements cannot be true?
 - $(x-z)^2 y$ is even
 - $(x-z) y^2$ is odd
 - $(x-z) y$ is odd
 - $(x-y)^2 z$ is even
- Pointing to a man in the photograph a lady said, "The father of his brother is the only son of my mother." How is this man in photograph related to the lady?
 - Brother
 - Son
 - Grandson
 - Nephew
- Find the odd number in the following series. 2, 9, 28, 65, 126, 216, 344, ...
 - 28
 - 65
 - 126
 - 216
- Average age of students of an adult school is 40 yr. 120 new students whose average age is 32 yr joined the school. As a result the average age is decreased by 4 yr. The number of students of the school after joining of the new student is
 - 1200
 - 120
 - 360
 - 240
- The letters P, Q, R, S, T, U and V not necessarily in that order represent seven consecutive integers from 22 to 33 and
 - U is as much less than Q as R is greater than S.
 - V is greater than U.
 - Q is the middle term.
 - P is greater than S.
 Then, the sequence of letters from the lowest value to the highest value, is
 - TVPQRSU
 - TRSQUPV
 - TUSQRPV
 - TVPQSRU
- The minimum number of tiles of size 16 by 24 required to form a square by placing them adjacent to one another is
 - 6
 - 8
 - 11
 - 16
- Five persons K, L, M, N and O are sitting around a dining table. K is the mother of M, M is actually the wife of O, N is the brother of K and L is the husband of K. How is N related to L?
 - Son
 - Cousin
 - Brother
 - Brother-in-law
- Three men A, B and C plays cards. If one loses the game he has to give ₹ 3. If he wins the game he will gain ₹ 3 each from the other two losers. If A has won 3 games, B loses ₹ 3, C wins ₹ 12, then the total number of games played is
 - 12
 - 21
 - 20
 - 6



Directions (Q. Nos. 17-19) Read the following passage carefully and answer the questions.

- A causes B or C but not both.
 - F occurs only if B occurs.
 - D occurs, if B or C occurs.
 - E occurs only if C occurs.
 - J occurs only if E or F occurs.
 - D causes G or H or both.
 - H occurs, if E occurs.
 - G occurs, if F occurs.
17. If A occurs, which may occurs?
- I. F and G
 - II. E and H
 - III. D
- (a) Only I
 - (b) Only II
 - (c) I and III or II and III, but not both
 - (d) I, II and III
18. If B occurs, which must occur?
- (a) D
 - (b) G
 - (c) H
 - (d) J
19. If J occurs, which must have occurred?
- (a) Both E and F
 - (b) Either B or C
 - (c) Both B and C
 - (d) None of these
20. If 'ROAST' is coded as 'PQYUR' in a certain language, then 'SLOPPY' is coded in that language as
- (a) MRNAQN
 - (b) NRMNQA
 - (c) QNMRNA
 - (d) RANNMO
21. If 'lelibroon' means 'yellow hat', 'plekafroti' means 'flower garden' and 'frotimix' means 'garden salad', then which word could mean 'yellow flower'?
- (a) Lelifroti
 - (b) Lelipleka
 - (c) Plekabroon
 - (d) Frotibroon
22. If + is *, - is +, * is / and / is -, then $6-9+8*3/20$ is equal to
- (a) -2
 - (b) 6
 - (c) 10
 - (d) 12
23. In a certain year, there were exactly four Fridays and four Mondays in January. On what day of the week did the 20th of January fall that year?
- (a) Saturday
 - (b) Sunday
 - (c) Thursday

- (d) Tuesday
24. Krishna said, "This girl is the wife of grandson of my mother". How is Krishna related to girl?
- (a) Father
 - (b) Father-in-law
 - (c) Husband
 - (d) Grandfather
25. Instead of walking along two adjacent sides of a rectangular field, a boy took a shortcut along the diagonal of the field and saved a distance equal to half the longer side. The ratio of the shorter side of the rectangle to the longer side is
- (a) $\frac{1}{2}$
 - (b) $\frac{2}{3}$
 - (c) $\frac{3}{4}$
 - (d) $\frac{4}{5}$
26. Each word in parenthesis below is formed in a method. This method is used in all four examples.
SNIP(NICE)PACE
TEAR(EAST)FAST
TRAY(RARE)FIRE
POUT(OURS)CARS
Based on this method, the word in the parenthesis of CANE(?)BATS is
- (a) NEAT
 - (b) CATS
 - (c) ANTS
 - (d) NETS
27. A study of native born residents in an area of Adivasis found that two-third of the children developed considerable levels of the nearsightedness after starting school, while their illiterate parents and grandparents, who had no opportunity for formal schooling, showed no signs of this disability. If the above statements are true, which of the following conclusions is most strongly supported by them?
- (a) Only people who have the opportunity for formal schooling develop nearsightedness
 - (b) People who are illiterate do not do not suffer from nearsightedness
 - (c) The nearsightedness in the children is caused by the visual stress required by reading and other class work
 - (d) Only literate people are nearsighted

Directions (Q.Nos.28-30) Read the following passage carefully and answer the questions.

- Five roommates Randy, Sally, Terry, Uma and Vernon each do one housekeeping taskmopping, sweeping, laundry, vacuuming or dusting one day a week, Monday through Friday.
- Vernon does not vacuum and does not do his task on Tuesday.
 - Sally does the dusting and does not do it on Monday or Friday.
 - The mopping is done on Thursday.
 - Terry does his task, which is not vacuuming, on Wednesday.
 - The laundry is done on Friday and not by Uma.
 - Randy does his task on Monday.
28. The task done by Terry on Wednesday is
- (a) Vacuuming





- (b) Dusting
(c) Mopping
(d) Sweeping

29. The day on which the vacuuming is done, is

- (a) Friday
(b) Monday
(c) Tuesday
(d) Wednesday

30. Sally does dusting on

- (a) Friday
(b) Monday
(c) Tuesday
(d) Wednesday

Direction(Q.Nos. 31-32) Read the following passage carefully and answer the questions.

P, Q, R, S, T, U, V and W are sitting round the circle and are facing the centre. P is second to the right of T, T is the neighbour of R and V. S is not the neighbour of P, V is the neighbour of U, Q is not between S and W and W is not between U and S.

31. Which two of the following are not neighbours?

- (a) RV
(b) UV
(c) RP
(d) QW

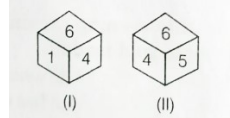
32. What is the position of S?

- (a) Between U and V
(b) Second to the right of P
(c) To the immediate right of W
(d) Data inadequate

33. The ratio between a two-digit number and the sum of the digits of that number is 4 : 1. If the digit in the unit's place is 3 more than the digit in ten's place, then the number is

- (a) 24
(b) 63
(c) 36
(d) 42

34. Two positions of a dice are shown below. When number 1 is in the top, What number will be at the bottom?



- (a) 2
(b) 3
(c) 5
(d) Cannot be determined

35. A, B, C, D, E, F and G are sitting in a line facing East. C is immediate to the right of D. B is at one of the extreme ends has E as his neighbour. G is between E and F. D is sitting third from the South end. Who is sitting third from North?

- (a) A
(b) E
(c) F
(d) G

36. There is a family party consisting of two fathers, two mothers, two sons, one father-in-law, one mother-in-law, one daughter-in-law, one grandfather, one grandmother and one grandson. What is the minimum number of persons required, so that this is possible?

- (a) 5

- (b) 6
(c) 7
(d) 8

37. If A is brother of B, C is brother of B and A is brother of D, then which of the following must be true?

- (a) A is brother of C
(b) B is brother of C
(c) D is brother of C
(d) B is brother of D

Directions(Q. Nos. 88-90) Read the following passage carefully and answer the questions.

Five houses lettered A, B, C, D and E are built in a row next to each other. The houses are lined up in the order A, B, C, D and E. Each of the five houses have coloured roofs and chimneys. The roof and chimney of each houses must be painted as follows.

- The roof must be painted either green, red or yellow.
- The chimney must be painted either white, black or red.
- No house may have the same colour chimney as the colour of roof.
- No house may use any of the same colours that adjacent house uses.
- House E has a green roof.
- House B has a red roof and a black chimney.

38. Which of the following is true?

- (a) Atleast two houses have a black chimney.
(b) Atleast two houses have red roofs.
(c) Atleast two houses have white chimneys
(d) Atleast two houses have green roofs

39. If house C has a yellow roof, then which of the following must be true?

- (a) House E has a white chimney
(b) House E has a black chimney
(c) House E has a red chimney
(d) House D has a red chimney

40. What is the maximum number of green roofs?

- (a) 1
(b) 2
(c) 3
(d) 4

ANSWER

1.	A	2.	A	3.	C	4.	B	5.	B
6.	d	7.	d	8.	c	9.	a	10.	c
11.	d	12.	b	13.	D	14.	C	15.	C
16.	B	17.	B	18.	D	19.	C	20.	A
21.	B	22.	A	23.	B	24.	C	25.	B
26.	D	27.	D	28.	B	29.	B	30.	A
31.	C	32.	D	33.	D	34.	B	35.	B
36.	b	37.	c	38.	d	39.	a	40.	a





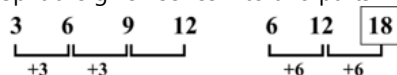
SOLUTION

1. (a) Let the distance covered by him is x km, then by condition,

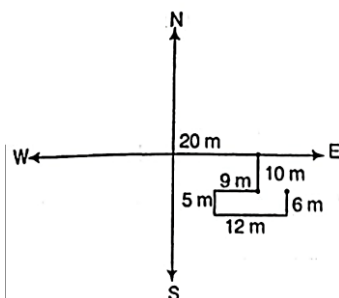
$$\begin{aligned} \frac{x}{4} - \frac{x}{5} &= \frac{12}{60} \\ \Rightarrow \frac{x}{20} &= \frac{1}{5} \\ \therefore x &= 4 \text{ km} \end{aligned}$$

2. (b) Given series, 3, 6, 6, 12, 9, 12

Split the given series into two parts



3. (a)



Hence, North direction is the man facing.

4. (b) Let x and y be the certain number of males and females.

Then, by condition,

$$x = \frac{1}{2}(y - 15)$$

$$\Rightarrow 2x = y - 15$$

$$\Rightarrow 2x - y = -15$$

$$5(x - 45) = y$$

$$5x - y = 225$$

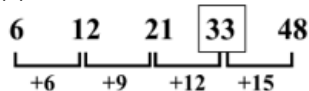
And On subtracting Eq. (i) from Eq. (ii), we get

$$3x = 240$$

$$\Rightarrow x = 80$$

$$\therefore \text{Number of males} = 80$$

5. (b)



Solution (Q.Nos. 6-8)

By condition DECAF (Shortest) $\downarrow$ (Longest)

6. (c) Between D and C

7. (d) C

8. (d) F

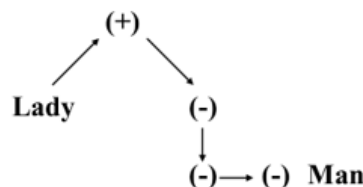
9. (a) x, y, z are distinct integers.
and x and y are odd positive integers and z is even positive integers.

Then, $(x - z) = \text{Odd number}$

$$(xc - z)^2 = \text{Odd positive number}$$

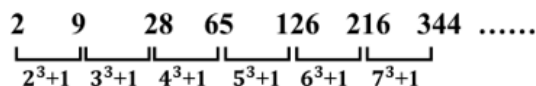
$$\text{and } (x - z)^2 \cdot y = \text{Odd} \times \text{Even} = \text{Even number}$$

10. (d)



So, man is nephew of the lady.

11. (d)



$$\begin{aligned} \text{or } 2 &= 1^3 + 1, & 9 &= 2^3 + 1, & 28 &= 3^3 + 1, \\ 65 &= 4^3 + 1, & 126 &= 5^3 + 1 \\ \text{and } 344 &= 7^3 + 1 \\ \text{But } 216 &= 6^3 + 0 \text{ which is odd number among them.} \end{aligned}$$

12. (d) Let the total number of students before joining new students = x .

After joining new 120 students = $x + 120$

Now, by condition,

$$x \times 40 + 120 \times 32 = (x + 120) \times 36$$

$$\Rightarrow 40x + 3840 = 36x + 4320$$

$$\Rightarrow 4x = 480$$

$$\therefore x = 120$$

$$\therefore \text{Total number of students} = x + 120 = 120 + 120 = 240$$

13. (c) By given condition, we get the required order (sequence) of letters from the lowest value to the highest value is

$$T < U < S < Q < R < P < V$$

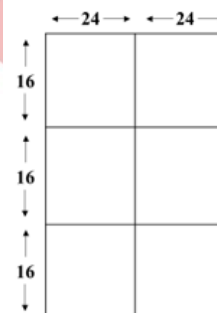
i.e., TUSQRPV

14. (a) From option (a),

$$\text{Let the number of tiles} = 6$$

$$\therefore \text{Total length} = 48$$

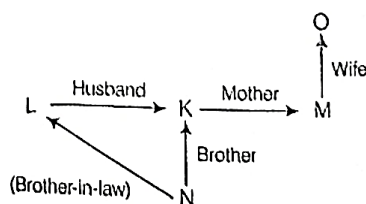
$$\text{and total breadth} = 48$$



Since, length = breadth

$$\therefore \text{Number of tiles form a square} = 6$$

15. (d)



16. (a) Required total number of games played is 12.





Solutions (Q.Nos. 17-19)

- (i) A causes B or C but not both.
- (ii) F occurs only if B occurs.
- (iii) D occurs if B or C occurs.
- (iv) E occurs only if C occurs.
- (v) C occurs only if E or F occurs
- (vi) D causes G or H or both
- (vii) H occurs if E occurs.
- (viii) G occurs if F occurs.

17. (c) From Statement (i), A causes B or C but not both. From Statement (ii), F occurs only if B occurs and from Statement (iii), D occurs if B or C occur. It means I and II may occur. From Statements (vi) and (vii), II and III are may occur. So, we conclude that I and III or II and III may occur but not both occur.
18. (b) From Statement (ii) that F occurs only if B occurs and from Statement (viii) that if G occurs if F occurs it means if B occurs G must occur.
19. (b) From Statement (v), that J occurs only if E or F occurs. From Statement (ii), F occurs only if B occurs and from Statement (iv), E occurs only if C occurs it means if J occurs either B or C must have occurs.

20. (c)

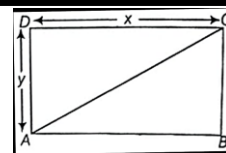
$$\begin{array}{ll} R \xrightarrow{-2} P & S \xrightarrow{-2} Q \\ O \xrightarrow{+2} P & L \xrightarrow{+2} N \end{array} \Rightarrow$$

$$\begin{array}{ll} A \xrightarrow{-2} Y & O \xrightarrow{-2} M \\ S \xrightarrow{+2} U & P \xrightarrow{+2} R \\ T \xrightarrow{-2} R & Y \xrightarrow{+2} A \end{array}$$

21. (b)
- lelibroon → yellow hat
pleka → flower garden
froti mix → garden salad
∴ Pleka, → flower
yellow → leli or broon
By option
yellow flower → lelipleka

22. (C) $E = 6 - 9 + 8 * \frac{3}{20}$
By given condition,
 $E = 6 + 9 * \frac{8}{3} - 20$
 $E = 6 + 3 * 8 - 20$
 $E = 6 + 24 - 20$
 $E = 6 + 4 = 10$

23. (b) Let in a month of January.
(4 times) Friday → 25, 18, 11, 4 (dates)
(4 times) Monday → 28, 21, 14, 7 (dates)
Then, required dates of Sunday,
Sunday → 27, 20, 13, 6
So, Sunday of the week did the 20th of January fall that year.
24. (b) Krishna is "father-in-law" of that girl.
25. (d) Let longer side = x = DC
and shorter side = y = AD



Now, by condition,
 $AC = y + \frac{x}{2}$

Now, In $\triangle ACD$,
 $AC^2 = AD^2 + CD^2$ (by Pythagoras theorem)

$$\begin{aligned} \Rightarrow \left(y + \frac{x}{2}\right)^2 &= y^2 + x^2 \\ \Rightarrow y^2 + \frac{x^2}{4} + xy &= y^2 + x^2 \\ \Rightarrow \frac{x^2}{4} + xy - x^2 &= 0 \\ \Rightarrow x\left\{\frac{x}{4} + y - x\right\} &= 0 \\ \Rightarrow x\left\{y - \frac{3x}{4}\right\} &= 0 \\ \Rightarrow \frac{y}{x} &= \frac{3}{4} \\ [\because x \neq 0] \end{aligned}$$

26. (c) SNIP (NICE) PACE
TEAR (EAST) FAST
TRAY (RARE) FIRE
POUT (OURS) CARS
∴ CANE (AN+TS) BATS
ANTS

27. (c) From the statements, we clearly say that the reason behind the nearsightedness of the children is caused by the visual stress required by reading and other class work.

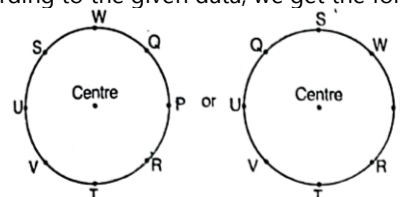
Solutions (Q.Nos. 28-30)

Randy	Vaccuming	Monday
Sally	Dusting	Tuesday
Terry	Sweeping	Wednesday
Uma	Mopping	Thursday
Vernon	Laundry	Friday

28. (d) Sweeping
29. (b) Monday
30. (c) Tuesday

Solutions (Q.Nos. 31-32)

According to the given data, we get the following figure



31. (a) R and V are not neighbours.
32. (d) The position of S is not fixed. So, data inadequate.
33. (c) Let the ten's place digit = x, then
By condition the unit place digit = x + 3
Now, according to question,
 $\frac{10x + (x + 3)}{4} = \frac{4}{7} \Rightarrow \frac{1 + x + 3}{2x + 3} = \frac{4}{7}$
 $\Rightarrow 1 + x + 3 = 8x + 12$
 $\Rightarrow 3x = 9$



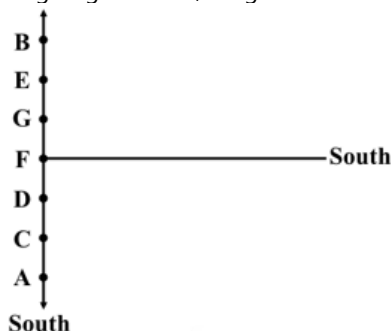


$$\Rightarrow x = 3$$

$$\therefore \text{Required number} = x(x + 3) = 3(3 + 3) = 36$$

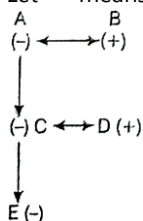
34. (c) After observation of given two dice, we get the number 5 is at the bottom of the dice, when number 1 is on the top.

35. (d) According to given data, we get the following figure



So, G is sitting third from North.

36. (a) Let '-' means 'male' and '+' means 'female'.

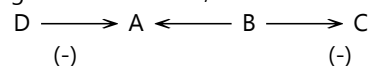


Two fathers (A, C)
Two sons (C, E)
One mother-in-law (B)
One grandfather (A)
One grandson (E)

Two mothers (B, D)
One father-in-law (A)
One daughter-in-law (D)
One grandmother (B)

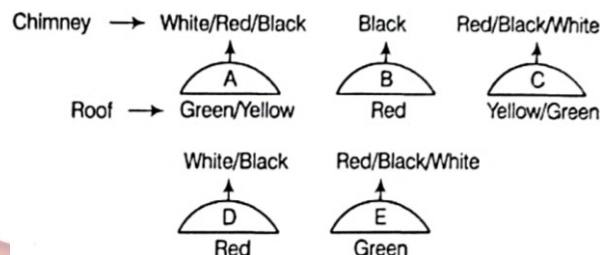
So, the minimum number of persons can be 5.

37. (a) According to the directions, the relation can be solved as



So, A is brother of C.

Solutions (Q.Nos. 88-90)



38. (b) From the above diagram, it is clear that atleast two houses have red roofs.

39. (a) If house C has a yellow roof it means house D has a red roof. So, house D never has a red chimney. So, chimney of D will be of black colour, so colour of chimney of house E will be white.

40. (c) The maximum number of green roofs are 3.

